

智能巡检车 PRO 版实验手册

雷达跟踪实验

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1

雷达跟踪讲解

1.1 程序功能说明

程序启动后，雷达扫描最近的一个物体，然后锁定，物体移动，小车跟着移动。

1.2 程序代码参考路径

进入 docker 容器后，该功能源码的位置位于，

```
/root/icar_ros2_ws/icar_ws/src/icar_laser/icar_laser/laser_Tracker_a1_X3.py
```

1.3 程序启动

1.3.1 启动命令

进入docker容器后，根据实际的车型和雷达型号，终端输入，

```
#启动小车底盘  
  
ros2 run icar_bringup Mcnamu_driver_X3  
  
#启动A1雷达  
  
ros2 launch slidar_ros2 slidar_launch.py  
  
#启动雷达跟踪程序  
  
ros2 run icar_laser laser_Tracker_a1_X3
```

1.3.2 查看话题通讯节点图

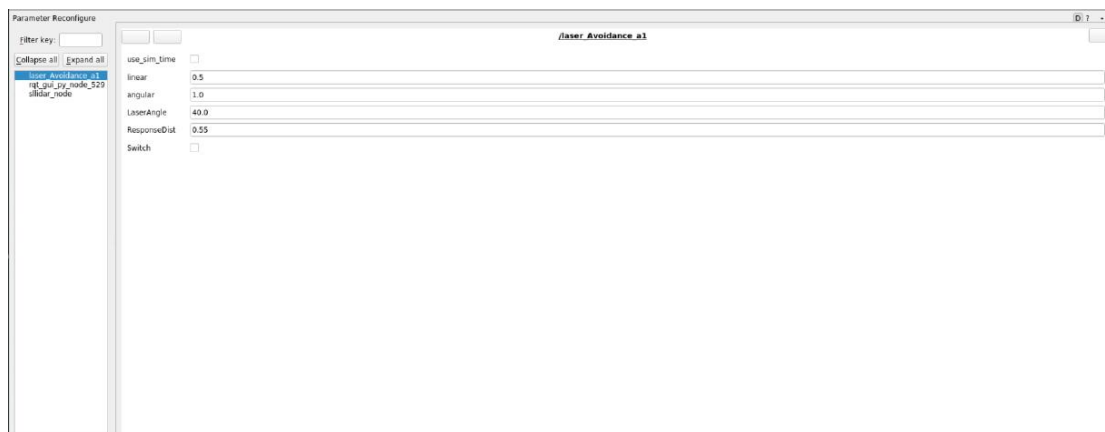
docker终端输入，

```
ros2 run rqt_graph rqt_graph
```



还可以通过动态参数调节器来设置参数的大小，终端输入，

```
ros2 run rqt_reconfigure rqt_reconfigure
```



各参数含义如下，

参数名称	参数含义
linear	线速度
angular	角速度
LaserAngle	雷达检测角度
ResponseDist	障碍物检测距离
Switch	玩法开关

以上参数都可调节，除了Switch以外，其他四个需要设置时候需要是小数，修改完，点击空白处才可以写入。

1.4 核心源码解析

主要看雷达的回调函数，这里说明了如何获取到每个角度的障碍物距离信息，

```
def registerScan(self, scan_data):
    if not isinstance(scan_data, LaserScan): return
    ranges = np.array(scan_data.ranges)
    self.Right_warning = 0
    self.Left_warning = 0
    self.front_warning = 0

    for i in range(len(ranges)):
        angle = (scan_data.angle_min + scan_data.angle_increment * i) * RAD2DEG#
        雷达的信息的angle是弧度制，这里要转换成角度进行计算
        if 160 > angle > 180 - self.LaserAngle:#angle是根据雷达的结构来设定判断范围的
            if ranges[i] < self.ResponseDist*1.5: #range[i]就是雷达扫描的结果，这里指
                得是距离信息
                self.Right_warning += 1
            if - 160 < angle < self.LaserAngle - 180:
                if ranges[i] < self.ResponseDist*1.5:

                    self.Left_warning += 1
        if abs(angle) > 160:
            if ranges[i] <= self.ResponseDist*1.5:
                self.front_warning += 1
        if self.Joy_active or self.Switch == True:
            if self.Moving == True:
                self.pub_vel.publish(Twist())
                self.Moving = not self.Moving
            return
        self.Moving = True
```

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雷达跟踪实验过程

1. 确认深度相机的编号。

```
jetson@jetson-desktop:~$ lsusb
Bus 002 Device 006: ID 0b95:1790 ASIX Electronics Corp. AX88179 Gigabit Ethernet
Bus 002 Device 005: ID 05e3:0626 Genesys Logic, Inc. USB3.1 Hub
Bus 002 Device 004: ID 05e3:0626 Genesys Logic, Inc.
Bus 002 Device 003: ID 2109:0817 VIA Labs, Inc.
Bus 002 Device 002: ID 2109:0822 VIA Labs, Inc. USB3.1 Hub
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 003: ID 0bda:c822 Realtek Semiconductor Corp. Bluetooth Radio
Bus 001 Device 013: ID 1a86:7523 OinHeng Electronics HL-340 USB-Serial adapter
Bus 001 Device 016: ID 2bc5:050f VIA Labs, Inc. USB2.0 Hub
Bus 001 Device 014: ID 2bc5:060f
Bus 001 Device 011: ID 05e3:0608 Genesys Logic, Inc. Hub
Bus 001 Device 015: ID 1c4f:0043 SiGma Micro Bluetooth Radio
Bus 001 Device 012: ID 30fa:0300 VIA Labs, Inc. USB2.0 Hub
Bus 001 Device 009: ID 05e3:0610 Genesys Logic, Inc. 4-port hub
Bus 001 Device 007: ID 05e3:0610 Genesys Logic, Inc. 4-port hub
Bus 001 Device 006: ID 0d8c:0012 C-Media Electronics, Inc. Bluetooth Radio
Bus 001 Device 010: ID 2109:0817 VIA Labs, Inc.
Bus 001 Device 008: ID 2207:0001 Fuzhou Rockchip Electronics Company
Bus 001 Device 005: ID 2109:2817 VIA Labs, Inc. USB2.0 Hub
Bus 001 Device 004: ID 10c4:ea60 Silicon Labs CP210x UART Bridge
Bus 001 Device 002: ID 2109:2822 VIA Labs, Inc. USB2.0 Hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
jetson@jetson-desktop:~$
```

观察设备编号有“2bc5”的两行，记录第四列的编号为 014、016（根据实际情况，编号有可能变化）。

异常处理：

如果只看到一行有“2bc5”，则需要重新拔插深度相机的 USB 连线，直到系统正确识别出现两行有“2bc5”。

2. 将上面的两个编号更新到~/run_docker.sh 文件中如下两行。

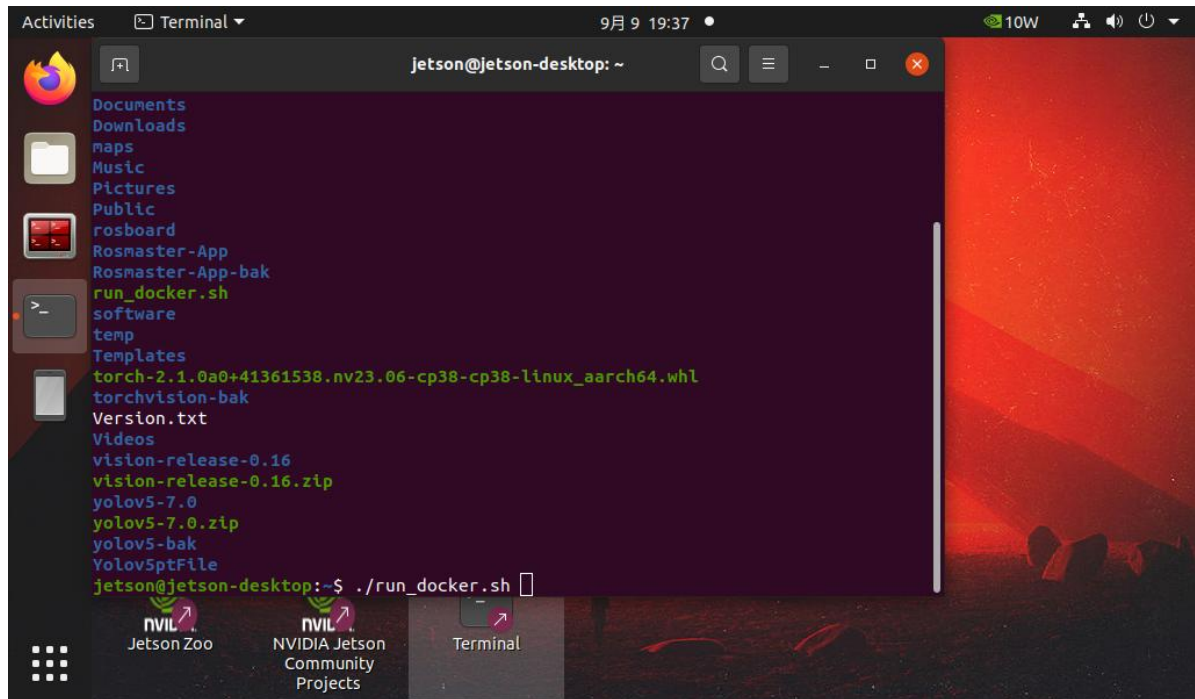
```
#!/bin/bash
xhost +

docker run -it \
--net=host \
--env="DISPLAY" \
--env="QT_X11_NO_MITSHM=1" \
-v /tmp/.X11-unix:/tmp/.X11-unix \
-v /home/jetson/temp:/root/icar_ros2_ws/temp \
-v /home/jetson/rosboard:/root/rosboard \
-v /home/jetson/maps:/root/maps \
-v /dev/bus/usb/001/014:/dev/bus/usb/001/014 \
-v /dev/bus/usb/001/016:/dev/bus/usb/001/016 \
--device=/dev/astradept \
--device=/dev/astrauvc \
--device=/dev/video0 \
--device=/dev/myserial \
--device=/dev/rplidar \
--device=/dev/input \
-p 9090:9090 \
-p 8888:8888 \
icar/ros-foxy:1.0.0 /bin/bash

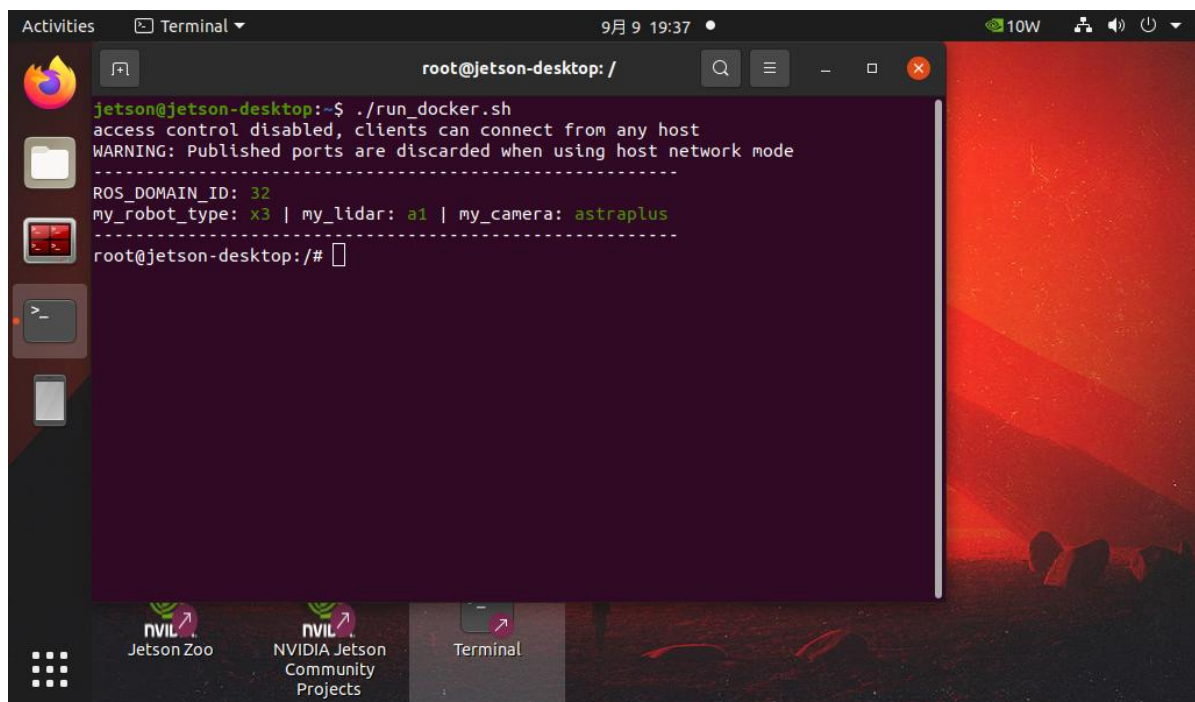
"run_docker.sh" 22L, 556C 22,15 All
```

保存退出。

3. 输入指令./run_docker.sh，启动 docker 镜像。



如无异常，出现如下信息，则说明进入 docker 镜像启动成功，并已进入 docker 环境。



4. 在新启动的 Terminal 窗口中,使用 `docker ps -a` 指令查看 `icar/ros-foxy:1.0.2` 镜像容器的 ID 前缀,比如: 这里的 4ca

```

jetson@jetson-desktop: ~/Rosmaster-App/rosmaster
data: ['\x00', '\x00', '\x00', '\x00', '\x00', '\x00', '\x15']
ext_type:11
data: ['\x18', '\x00', '\xf5', '\xff', '\xd9', '\xff', '\x28', '\x4', '\x8', '\x4', '\xe4', '\x40', '\xef', '\x1', '\xf3', '\xfe', '\x01', '\xf7', '\x99']
ext_type:10
data: ['\x00', '\x00', '\x00', '\x00', '\x00', '\x00', '\x71', '\x85']
ext_type:13
data: ['\xa6', '\xe6', '\xff', '\xff', '\x5c', '\xd', '\x0', '\x0', '\x78', '\x2', '\x0', '\x0', '\xc9', '\xe5', '\xff', '\xff', '\x39']
ext_type:12
data: ['\x00', '\x00', '\x00', '\x00', '\x00', '\x00', '\x15']
ext_type:11
data: ['\x19', '\x00', '\xf5', '\xff', '\xd9', '\xff', '\x64', '\x4', '\xf8', '\x3', '\x18', '\x43', '\xf5', '\x1', '\xef', '\xfe', '\x06', '\xf7', '\x3']
Exception in thread task_serial_receive:
Traceback (most recent call last):
  File "/usr/lib/python3.8/threading.py", line 932, in _bootstrap_inner
    self.run()
  File "/usr/lib/python3.8/threading.py", line 870, in run
    self._target(*self._args, **self._kwargs)
  File "/usr/local/lib/python3.8/dist-packages/Rosmaster_Lib-3.3.1-py3.8.egg/Rosmaster_Lib/Rosmaster_Lib.py", line 267, in __receive_data
    File "/usr/local/lib/python3.8/dist-packages/serial/serialposix.py", line 595, in read
        raise SerialException(
serial.serialutil.SerialException: device reports readiness to read but returned no data (device disconnected or multiple access on port?)
[255, 252, 4, 54, 1, 59]
follow the line: ['\xff', '\xfc', '\x4', '\x36', '\x1', '\x3b']
[255, 252, 4, 54, 0, 58]
follow the line: ['\xff', '\xfc', '\x4', '\x36', '\x0', '\x3a']
jetson@jetson-desktop:~/Rosmaster-App/rosmaster$ docker ps -a
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS              NAMES
4ca101a51e0c       lcar/ros-foxy:1.0.2  "/bin/bash"        22 hours ago        Exited (0) 2 hours ago              pensive_curle
7768d44a5623       yahboomtechnology/ros-foxy:5.0.1  "/bin/bash"        23 hours ago        Exited (0) 2 hours ago              stoic_chatelet
jetson@jetson-desktop:~/Rosmaster-App/rosmaster$

```

5. 在 Terminal 窗口中输入 cd 指令进入 HOME 目录:

```

jetson@jetson-desktop: ~
ext_type:11
data: ['\x18', '\x00', '\xf5', '\xff', '\xd9', '\xff', '\x28', '\x4', '\x8', '\x4', '\xe4', '\x40', '\xef', '\x1', '\xf3', '\xfe', '\x01', '\xf7', '\x99']
ext_type:10
data: ['\x00', '\x00', '\x00', '\x00', '\x00', '\x00', '\x71', '\x85']
ext_type:13
data: ['\xa6', '\xe6', '\xff', '\xff', '\x5c', '\xd', '\x0', '\x0', '\x78', '\x2', '\x0', '\x0', '\xc9', '\xe5', '\xff', '\xff', '\x39']
ext_type:12
data: ['\x00', '\x00', '\x00', '\x00', '\x00', '\x00', '\x15']
ext_type:11
data: ['\x19', '\x00', '\xf5', '\xff', '\xd9', '\xff', '\x64', '\x4', '\xf8', '\x3', '\x18', '\x43', '\xf5', '\x1', '\xef', '\xfe', '\x06', '\xf7', '\x3']
Exception in thread task_serial_receive:
Traceback (most recent call last):
  File "/usr/lib/python3.8/threading.py", line 932, in _bootstrap_inner
    self.run()
  File "/usr/lib/python3.8/threading.py", line 870, in run
    self._target(*self._args, **self._kwargs)
  File "/usr/local/lib/python3.8/dist-packages/Rosmaster_Lib-3.3.1-py3.8.egg/Rosmaster_Lib/Rosmaster_Lib.py", line 267, in __receive_data
    File "/usr/local/lib/python3.8/dist-packages/serial/serialposix.py", line 595, in read
        raise SerialException(
serial.serialutil.SerialException: device reports readiness to read but returned no data (device disconnected or multiple access on port?)
[255, 252, 4, 54, 1, 59]
follow the line: ['\xff', '\xfc', '\x4', '\x36', '\x1', '\x3b']
[255, 252, 4, 54, 0, 58]
follow the line: ['\xff', '\xfc', '\x4', '\x36', '\x0', '\x3a']
jetson@jetson-desktop:~/Rosmaster-App/rosmaster$ docker ps -a
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS              NAMES
4ca101a51e0c       lcar/ros-foxy:1.0.2  "/bin/bash"        22 hours ago        Exited (0) 2 hours ago              pensive_curle
7768d44a5623       yahboomtechnology/ros-foxy:5.0.1  "/bin/bash"        23 hours ago        Exited (0) 2 hours ago              stoic_chatelet
jetson@jetson-desktop:~/Rosmaster-App/rosmaster$ cd
jetson@jetson-desktop:~$ vi .bashrc

```

6. 使用 vi .bashrc 编辑环境变量文件，定义 is、id、it 三条快捷指令:


```

. /usr/share/bash-completion/bash_completion
elif [ -f /etc/bash_completion ]; then
. /etc/bash_completion
fi
alias python=python3
ip=$(ip addr show wlan0 | grep -o 'inet [0-9]\+\.[0-9]\+\.[0-9]\+\.[0-9]\+' | grep -o [0-9].*)
if [ -z $ip ]; then
ip=$(ip addr show lo | grep -o 'inet [0-9]\+\.[0-9]\+\.[0-9]\+\.[0-9]\+' | grep -o [0-9].*)
fi
echo "-----"
echo -e "MY_IP: \033[32m$ip\033[0m"
echo "-----"

export PATH=/usr/local/cuda-11.4/bin${PATH:+:${PATH}}
export LD_LIBRARY_PATH=/usr/local/cuda-11.4/lib64${LD_LIBRARY_PATH:+:${LD_LIBRARY_PATH}}
export CUDA_ROOT=/usr/local/cuda

alias ros='cd /home/jetson/Rosmaster-App/rosmaster'
alias run='/home/jetson/Rosmaster-App/rosmaster/start_app.sh'
alias start='python rosmaster_pid.py 1 50 40 2000'
alias stop='python rosmaster_pid.py 7 50 40 2000'
alias app='cd /home/jetson/Rosmaster-App/rosmaster;python3 app_sim_run.py'
alias back='if [ ! -d ~/valid ]; then mkdir -p ~/valid;fi;cp -r /Rosmaster-App/rosmaster/license.json ~/valid;cp -r /Rosmaster-App/rosmaster/enc.txt ~/valid'
alias restore='cp -r /valid/license.json ~/Rosmaster-App/rosmaster;cp -r /valid/enc.txt ~/Rosmaster-App/rosmaster/'

alias s='docker start 7768'
alias t='docker stop 7768'
alias d='docker exec -it 7768 /bin/bash'

alias ts='docker start 4ca'
alias it='docker stop 4ca'
alias id='docker exec -it 4ca /bin/bash'
.bashrc 146L, 5048C
  
```

编辑完成后保存退出。

在 Terminal 窗口运行指令 `source .bashrc` 指令生效环境变量文件

此时，在 Terminal 窗口中可以通过 s、t、d 三个快捷指令实现对自动导航容器的启动、停止和登入操作：

快捷指令	完整指令	指令用途
is	<code>docker start 2169</code>	启动 ID 前缀为 2169 的 docker 容器
it	<code>docker stop 2169</code>	停止 ID 前缀为 2169 的 docker 容器
id	<code>docker exec -it 2169 /bin/bash</code>	登入 ID 前缀为 2169 的 docker 容器

7. 在 Terminal 窗口中运行快捷指令 id，进入 icar 镜像的 docker 容器：

```

jetson@jetson-desktop:~$ t
7768
jetson@jetson-desktop:~$ ls
alias.txt          Pictures           temp
Archiconda3-0.2.3-Linux-aarch64.sh Public            redhsv.py
cnd.txt            Rosmaster-App    rosboard
code              Rosmaster-App-bak  torchvision-bak
Desktop           Rosmaster-App-bak  trackObj.py
Documents         run_docker_guodrive.sh  Verston.txt
Downloads         run_docker.sh      vision-release-0.16
findObj.py        Samples-guide.pdf  yolov5-7.0
maps             software           Yolov5ptFile
Music
jetson@jetson-desktop:~$ ts
4ca
jetson@jetson-desktop:~$ id
-----
ROS_DOMAIN_ID: 32
my_robot_type: x3 | my_lidar: a1 | my_camera: astraplus
-----
root@jetson-desktop:/#
  
```

iCar 镜像 docker 容器中，定义了如下快捷指令：

快捷指令	完整指令	指令用途
------	------	------

n1	ros2 run icar_bringup Mcnamu_driver_X3	启动小车底盘驱动 ROS 应用节点
n2	ros2 launch sllidar_ros2 sllidar_launch.py	启动小车雷达 ROS 应用节点
n3	ros2 run icar_laser laser_Avoidance_a1_X3	启动小车雷达避障 ROS 应用节点
n4	ros2 run icar_laser laser_Tracker_a1_X3	启动雷达跟随 ROS 应用节点
n5	ros2 run icar_laser laser_Warning_a1_X3	启动雷达警卫 ROS 应用节点
m1	ros2 launch astra_camera astra_launch.xml	启动深度相机 ROS 应用节点
m2	ros2 run icar_astra colorHSV	启动 HSV 颜色分析 ROS 应用节点
m3	ros2 run icar_astra colorTracker	启动颜色追踪 ROS 应用节点

8. 在 iCar 镜像的 docker 容器中执行快捷指令 n1

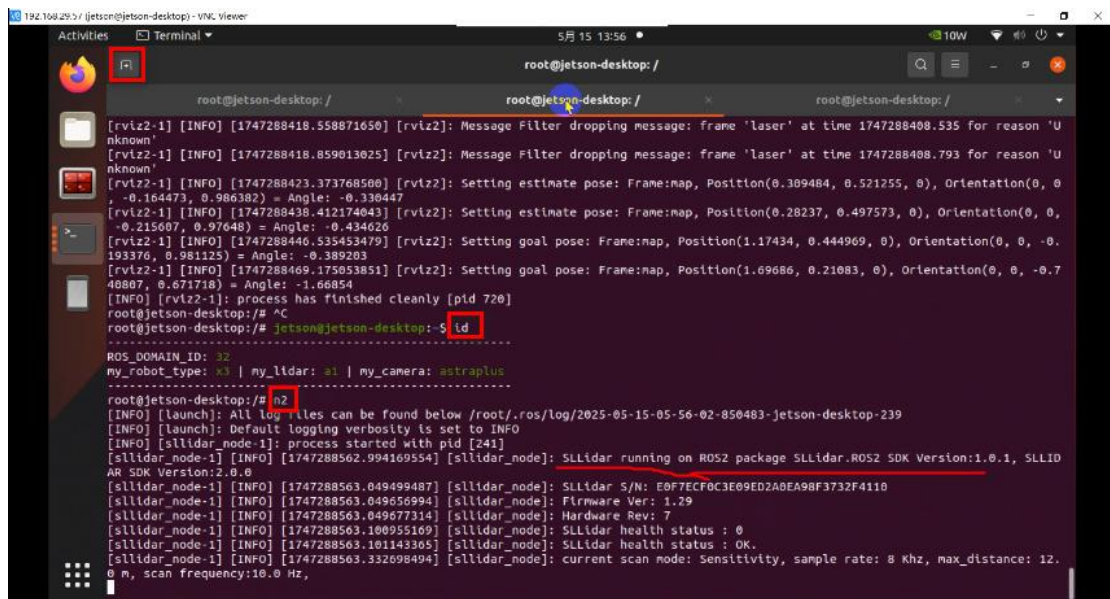
```

root@jetson-desktop: /
[INFO] [sllidar_node-8]: process has finished cleanly [pid 615]
root@jetson-desktop:/# exit
exit
jetson@jetson-desktop:~$ t
7760
jetson@jetson-desktop:~$ ls
alias.txt          Pictures           temp
Archiconda3-0.2.3-Linux-aarch64.sh Public            Templates
cnd.txt            redhsv.py         torch-2.1.0a0+41361938.nv23.06-cp38-cp38-linux_aarch64.whl
code              rosboard          torchvision-bak
Desktop           Rosmaster-App     trackObj.py
Documents        Rosmaster-App-bak Version.txt
Downloads        run_docker_autodrive.sh Videos
flndObj.py       run_docker.sh     vision-release-0.16
maps             Samples-guide.pdf yolov5-7.0
Music            software           Yolov5ptFile
jetson@jetson-desktop:~$ is
4ca
jetson@jetson-desktop:~$ id
-----
ROS_DOMAIN_ID: 32
my_robot_type: X3 | my_lidar: a1 | my_camera: astrapius
-----
root@jetson-desktop:/# n1
Rosmaster Serial Opened! Baudrate=115200
X3
Inu_Link
1.0
1.0
5.0
-----create receive threading-----

```

可以看到小车底盘驱动 ROS 应用节点被启动。

9. 再启动一个 Terminal 窗口，运行指令 id 进入 iCa 镜像容器，并执行快捷指令 n2:



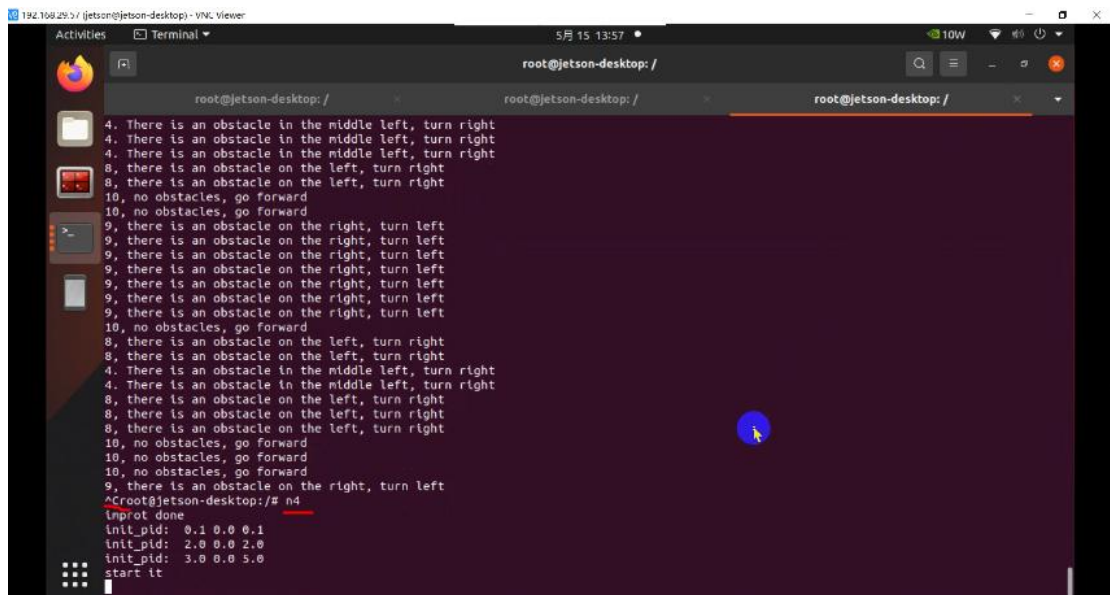
```

root@jetson-desktop: /
[rviz2-1] [INFO] [1747288418.558871650] [rviz2]: Message Filter dropping message: frame 'laser' at time 1747288408.535 for reason 'Unknown'
[rviz2-1] [INFO] [1747288418.859013025] [rviz2]: Message Filter dropping message: frame 'laser' at time 1747288408.793 for reason 'Unknown'
[rviz2-1] [INFO] [1747288423.373768500] [rviz2]: Setting estimate pose: Frame:map, Position(0.389484, 0.521255, 0), Orientation(0, 0, -0.164473, 0.986382) = Angle: -0.330447
[rviz2-1] [INFO] [1747288438.412174043] [rviz2]: Setting estimate pose: Frame:map, Position(0.28237, 0.497573, 0), Orientation(0, 0, -0.215607, 0.97648) = Angle: -0.434626
[rviz2-1] [INFO] [1747288446.535453479] [rviz2]: Setting goal pose: Frame:map, Position(1.17434, 0.444969, 0), Orientation(0, 0, -0.193376, 0.981125) = Angle: -0.389203
[rviz2-1] [INFO] [1747288469.175053851] [rviz2]: Setting goal pose: Frame:map, Position(1.69686, 0.21083, 0), Orientation(0, 0, -0.740887, 0.671718) = Angle: -1.66854
[INFO] [rviz2-1]: process has finished cleanly [pid 720]
root@jetson-desktop: /# ^C
root@jetson-desktop: /# jetson@jetson-desktop:~$ id
-----
ROS_DOMAIN_ID: 32
my_robot_type: x3 | my_lidar: a1 | my_camera: astraplus
-----
root@jetson-desktop: /# n2
[INFO] [launch]: All log files can be found below /root/.ros/log/2025-05-15-05-56-02-850483-jetson-desktop-239
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [sllidar_node-1]: process started with pid [241]
[sllidar_node-1] [INFO] [1747288562.994169554] [sllidar_node]: SLLidar running on ROS2 package SLLidar.ROS2 SDK Version:1.0.1, SLLIDAR SDK Version:2.0.0
[sllidar_node-1] [INFO] [1747288563.049499487] [sllidar_node]: SLLidar S/N: E0F7ECF0C3E09E02A0EA98F3732F4110
[sllidar_node-1] [INFO] [1747288563.049656994] [sllidar_node]: SLLidar Firmware Ver: 1.29
[sllidar_node-1] [INFO] [1747288563.049677314] [sllidar_node]: SLLidar Hardware Rev: 7
[sllidar_node-1] [INFO] [1747288563.100955169] [sllidar_node]: SLLidar health status : 0
[sllidar_node-1] [INFO] [1747288563.101143365] [sllidar_node]: SLLidar health status : OK.
[sllidar_node-1] [INFO] [1747288563.332698494] [sllidar_node]: current scan mode: Sensitivity, sample rate: 8 KHz, max_distance: 12.0 m, scan frequency:10.0 Hz,

```

可以看到雷达驱动节点被启动。

- 再启动一个 Terminal 窗口，运行指令 `id` 进入 iCa 镜像容器，并执行快捷指令 `n4`：启动雷达自动跟随指令：



```

4. There is an obstacle in the middle left, turn right
4. There is an obstacle in the middle left, turn right
4. There is an obstacle in the middle left, turn right
8. there is an obstacle on the left, turn right
8. there is an obstacle on the left, turn right
10. no obstacles, go forward
10. no obstacles, go forward
9. there is an obstacle on the right, turn left
9. there is an obstacle on the right, turn left
9. there is an obstacle on the right, turn left
9. there is an obstacle on the right, turn left
9. there is an obstacle on the right, turn left
9. there is an obstacle on the right, turn left
10. no obstacles, go forward
8. there is an obstacle on the left, turn right
8. there is an obstacle on the left, turn right
4. There is an obstacle in the middle left, turn right
4. There is an obstacle in the middle left, turn right
8. there is an obstacle on the left, turn right
8. there is an obstacle on the left, turn right
8. there is an obstacle on the left, turn right
10. no obstacles, go forward
10. no obstacles, go forward
10. no obstacles, go forward
9. there is an obstacle on the right, turn left
n4
n4 done
init_pid: 0.1 0.0 0.1
init_pid: 2.0 0.0 2.0
init_pid: 3.0 0.0 5.0
start it

```

小车启动了雷达跟随 ROS 应用节点。

- 人在小车前面走动，小车会跟随人进行前进或者后退以保持一个距离

